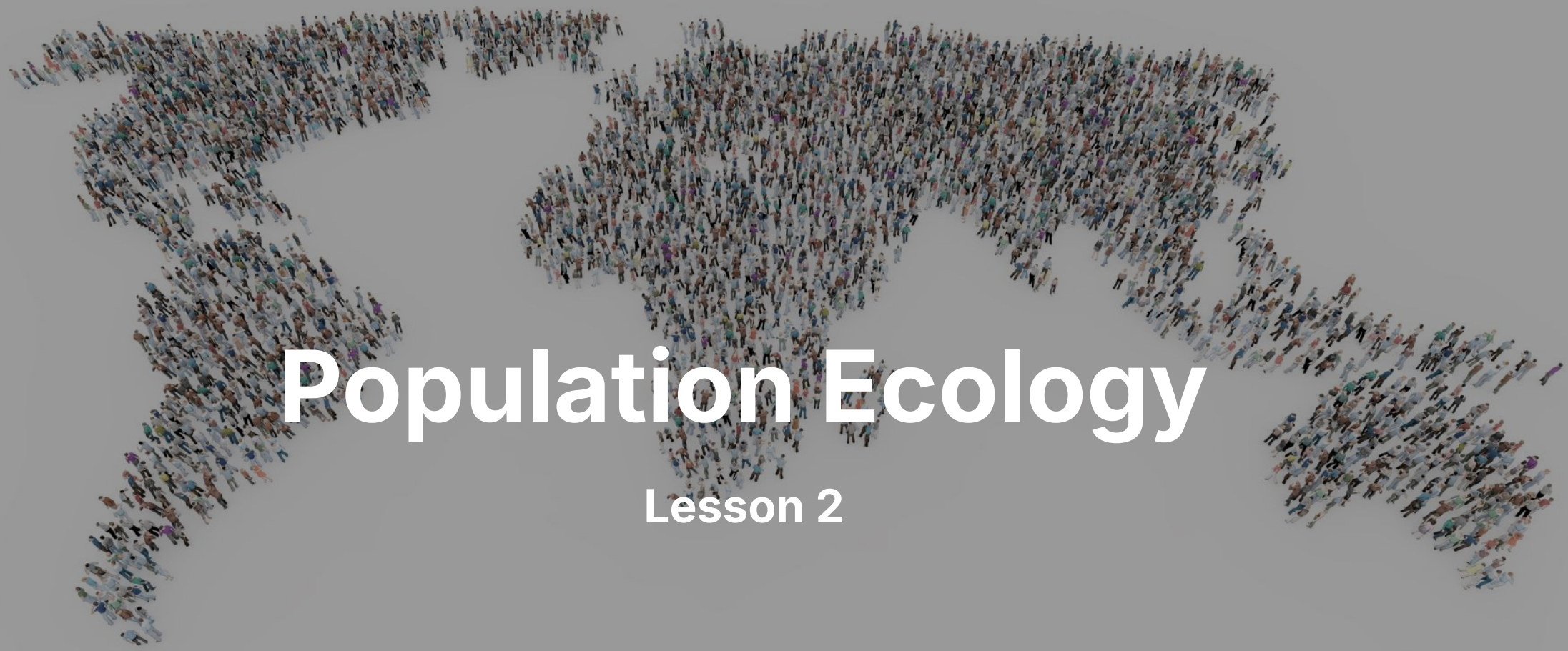


Population Ecology

Lesson 2

Population Ecology

I. Population Dynamics
II. Human Populations

Learning objectives

At the end of this lesson, students should be able to:

1. Describe how ecologists measure population size and density.
2. Describe three different patterns of population distribution.
3. Explain how the carrying capacity of a habitat may change.
4. Describe human population patterns and trends.

POPULATION ECOLOGY

- **Population Ecology** is the study of how **populations** of organisms **interact** with their **environment** over time.

Key Population Characteristics

Population Size (N)

Population Density

Population Distribution

- Clumped
- Uniform
- Random

POPULATION SIZE (N)

- Population size refers to the **total number of individuals** of a species in each area at a specific time.
- It gives a basic idea of how large or small a population is.

Examples:

- A forest has **200 deer** → Population size = 200
- A classroom has **40 students** → Population size = 40
- A city has **1 million people** → Population size = 1,000,000

Estimating population size

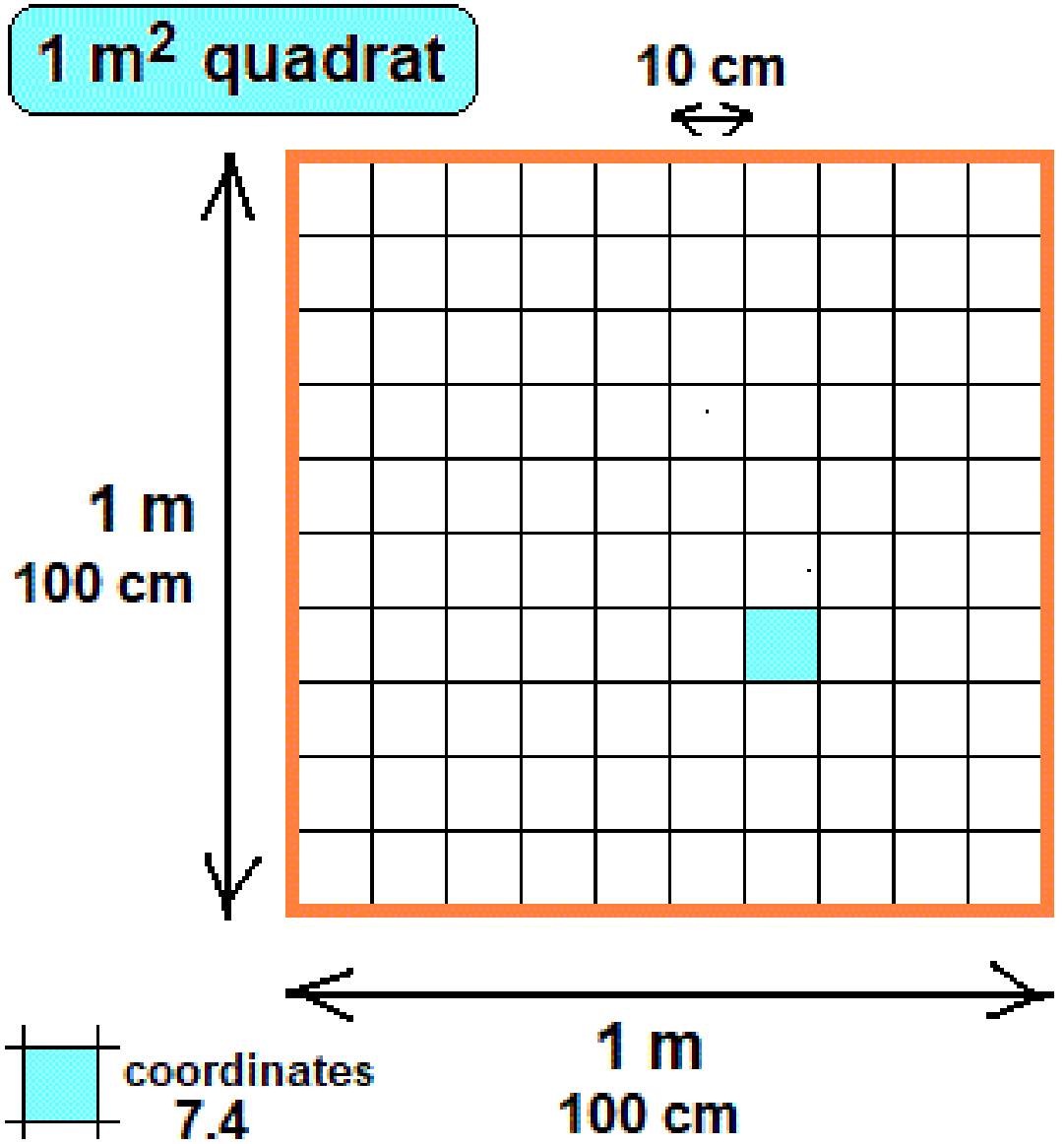
1. Quadrat method

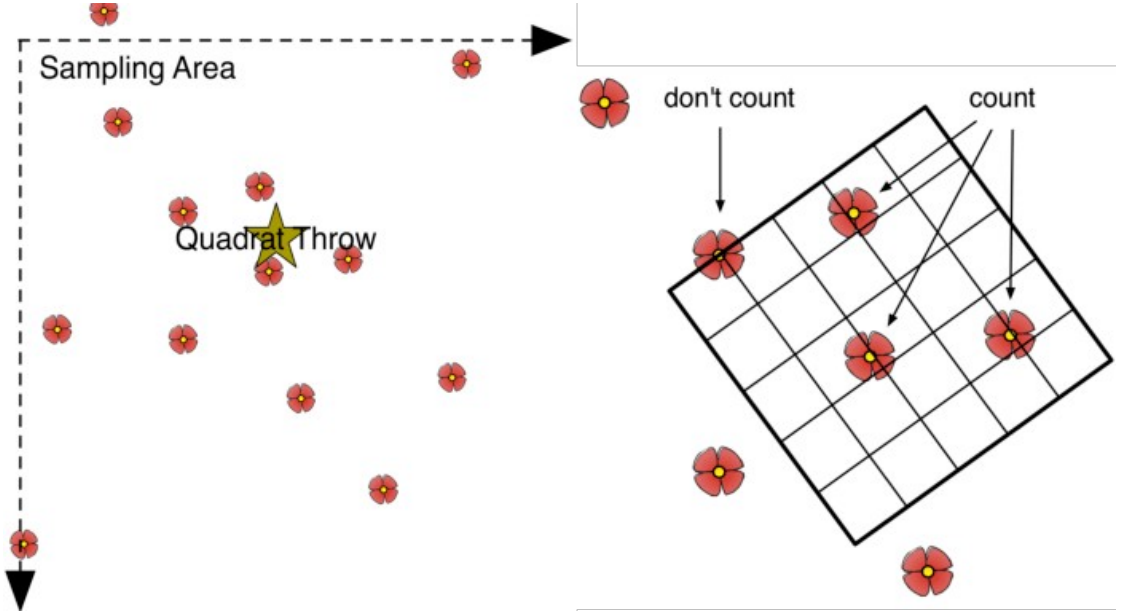
- Used for **immobile or slow-moving organisms** (e.g., plants)
- Quadrat (square frame) placed **randomly** in the habitat.
- Count individuals inside each quadrat.
- Repeat many times for **accuracy.**

Formula:

- Population = *Average per quadrat* × *Total number of quadrats*

Quadrat method




Sampling Area
Quadrat Throw
don't count
count



Example:

- A researcher wants to estimate the number of **grass plants** in a field.
- A **1 m × 1 m quadrat** is placed randomly on the ground **10 times**.
- The number of plants counted in each quadrat:
- 12, 15, 10, 13, 14, 11, 16, 12, 13, 14

Step 1: Find the average

- Total = 130 plants
- Average = 130 ÷ 10
  = **13 plants per m²**

Step 2: Estimate total population

- If the field is **100 m²**:
- Estimated population = 13 × 100 = **1300 plants**

- About **1300 plants** are estimated in the whole field based on the sample.

Cont.

2. Mark–recapture method

- Used for **mobile organisms** (e.g., mammals)
- First sample: capture, **mark**, and release animals
- Second sample: capture again and count **marked individuals**
- Population estimated using formula:
- $$N = \frac{marked first\ catch\ x\ total\ second\ catch}{marked\ second\ catch}$$

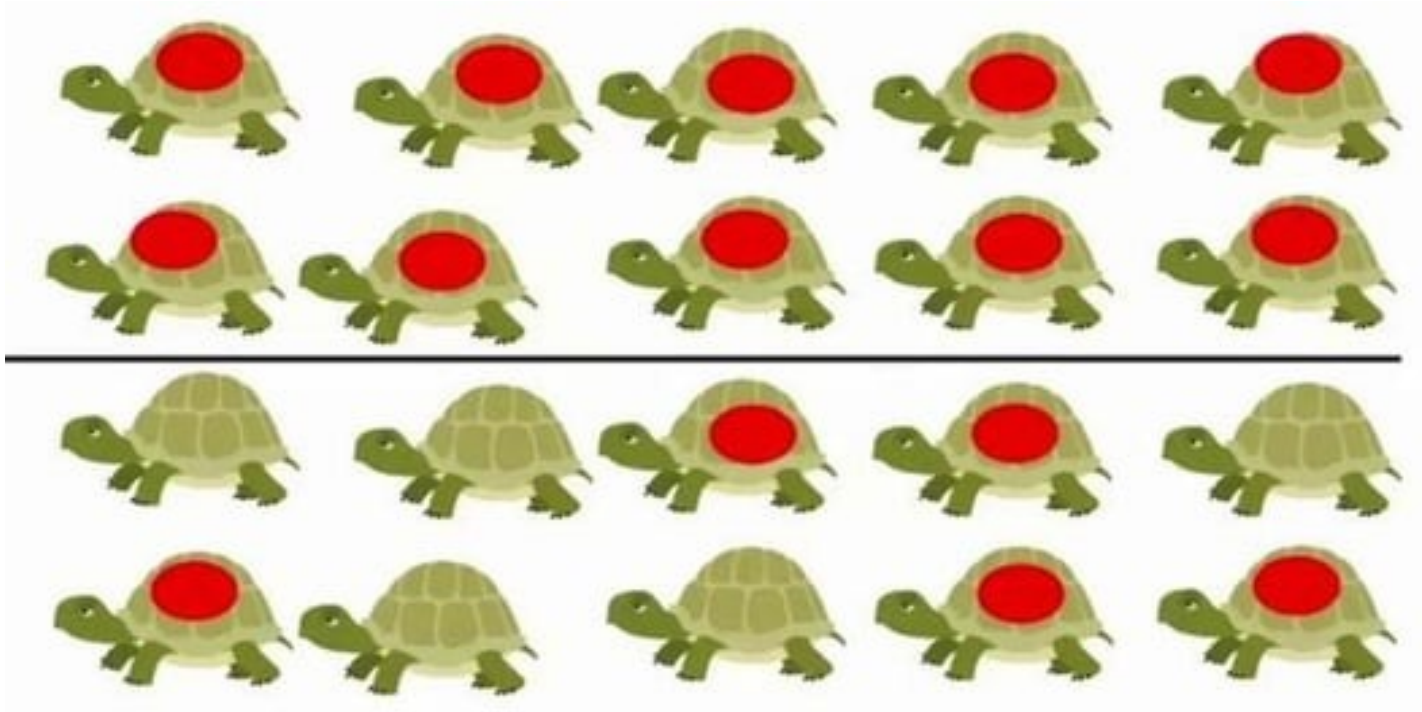
Key considerations

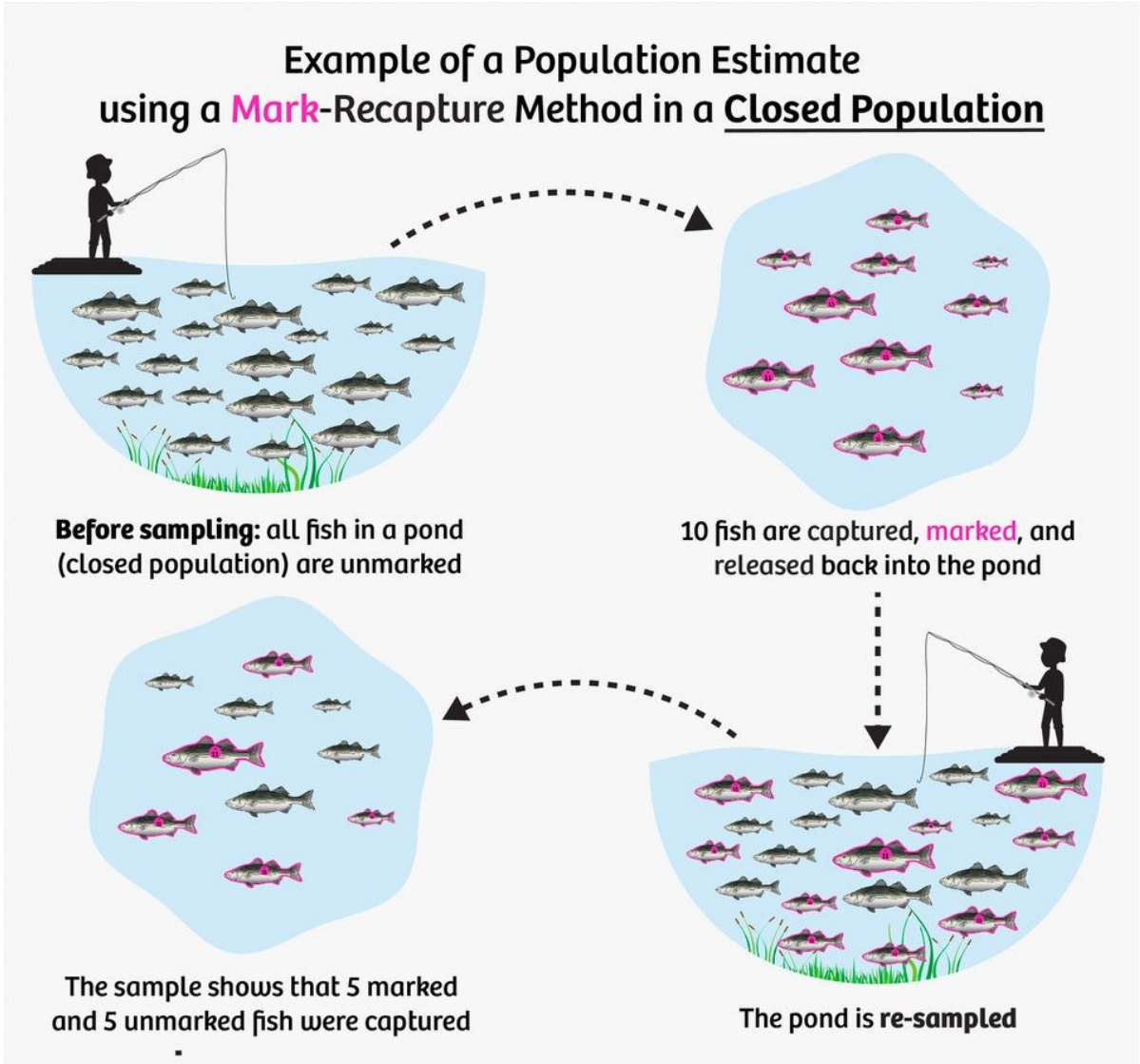
- Results are **estimates, not exact.**
- Accuracy depends on:
  - Random sampling.
  - Proper mixing of marked animals.
  - No loss of marks.
  - Stable population (no births, deaths, migration).

Formula

- $$N = \frac{marked first\ catch\ x\ total\ second\ catch}{marked\ second\ catch}$$

Mark–recapture method





$$N = \frac{marked first\ catch\ x\ total\ second\ catch}{marked\ second\ catch}$$

$$N = \frac{10\ x\ 10}{5}$$

$$N = 20$$

Try this!

Example:

- If 80 field mice are captured, marked, and released into the forest, then a second trapping of 100 field mice is captured, and 20 of them are marked. What is the population of the field mouse in that area?

- $N = \frac{marked first\ catch\ x\ total\ second\ catch}{marked\ second\ catch}$
- $N = \frac{80\ x\ 100}{20}$
- $N = 400$

POPULATION DENSITY

- Population density is the **number of individuals per unit area or volume**.

Formula:

- $Population\ Density = \frac{Number\ of\ Individuals}{Area\ or\ Volume}$

- It shows **how crowded or spread-out** individuals are.
  - High density → more competition for food, water, and space
  - Low density → individuals are more spread out

Examples:

- 100 plants in 1 square meter → **High density**
- 100 plants in 1 hectare → **Low density**

Examples:

- **Urban areas (cities)** → high population density.

- **Deserts** → low population density

Why it matters:

- High density can lead to
  - Disease spread
  - Competition
- Low density can make
  - Finding mates difficult.

POPULATION DISTRIBUTION (DISPERSION)

- Population distribution refers to **how individuals are spaced within a given area**.

Three main types:

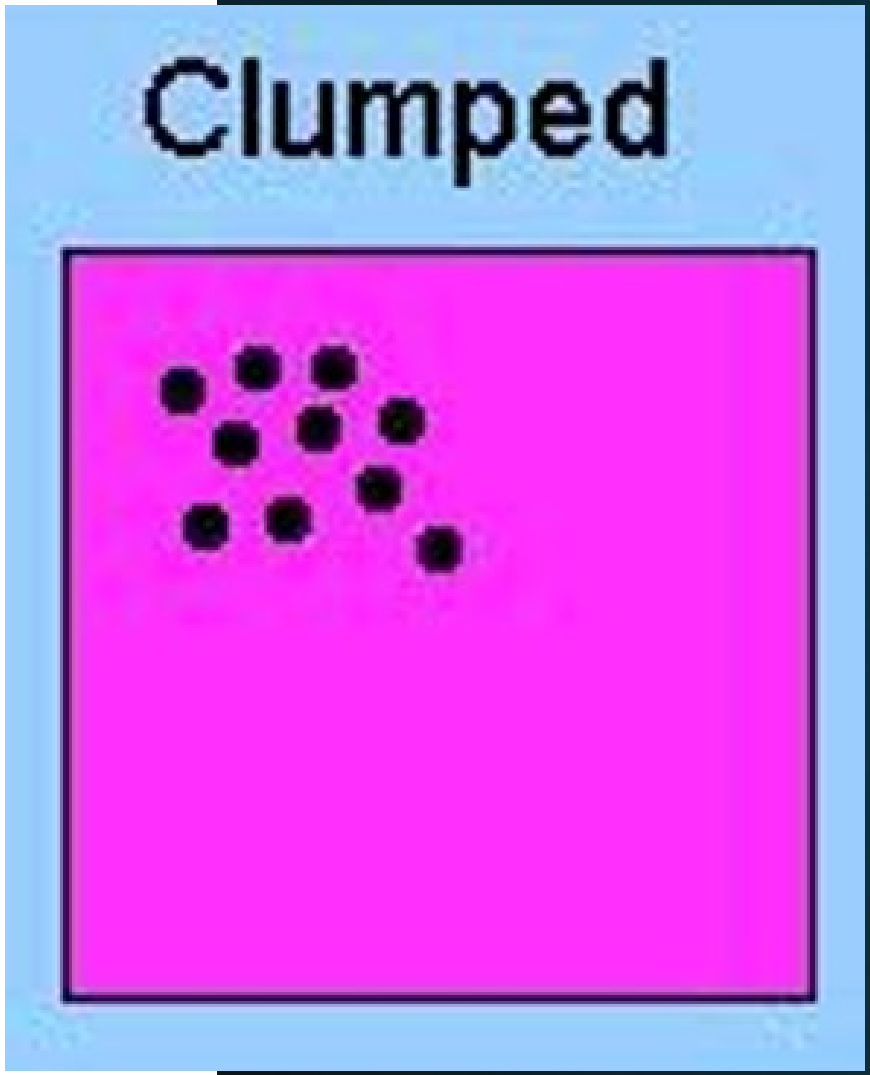
1. **Clumped Distribution**
2. **Uniform Distribution**
3. **Random Distribution**

1. Clumped Distribution

Individuals are grouped in patches.

- **Why did it happen?**
  - Resources are concentrated in certain areas.
  - Protection from predators.
  - Social behavior (living in groups).

The **most common** type of distribution in nature.



Examples

1. African Elephants

- Elephants gather in groups around **rivers and watering holes**, especially during dry seasons.

Reason:

- Water is limited and found only in certain places.



2. Schools of Fish

- Fish such as **sardines or tuna** move in large groups in the ocean.

Reason:

- Protection from predators.
- Easier to find food.



Examples

3. Human Populations (Cities)

- People live **close together in cities** like Manila, Tokyo, or New York.

Reason:

- Jobs
- Access to services (schools, hospitals)
- Transportation

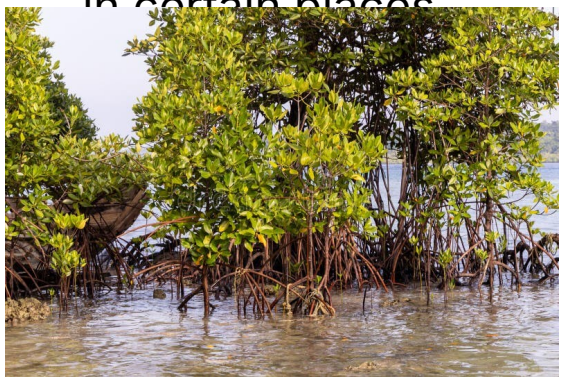


4. Trees in Fertile Areas

- Trees often grow in **clusters where soil and water are favorable**.
- **Example:** Mangrove forests along coastal areas.

Reason:

- Suitable environmental conditions exist only in certain places



Examples

5. Ant Colonies

- Ants live together in **colonies or nests**.

Reason:

  - Social behavior and cooperation

- Example: Anthills found in yards or forests.



6. Penguin Colonies

- Penguins gather in large groups during breeding season.
- **Reason:**
  - Protection against predators.
  - Warmth in cold environments.
- Example: Emperor penguins in Antarctica.



Examples

7. Grazing Animals (Cows, Zebras, Wildebeest)

- Herbivores gather in **herds** in grasslands.
- **Reason:**
  - Safety in numbers
  - Availability of grass
- Example: Wildebeest migration in Africa



8. Bird Nesting Colonies

- Birds like **herons or seagulls** nest close together.
- **Reason:**
  - Protection and breeding advantage
- Example: Seagulls nesting on cliffs



Example

9. Mushrooms After Rain

- Mushrooms often grow in **clusters in moist areas**.

Reason:

- Favorable conditions (moisture, organic matter)

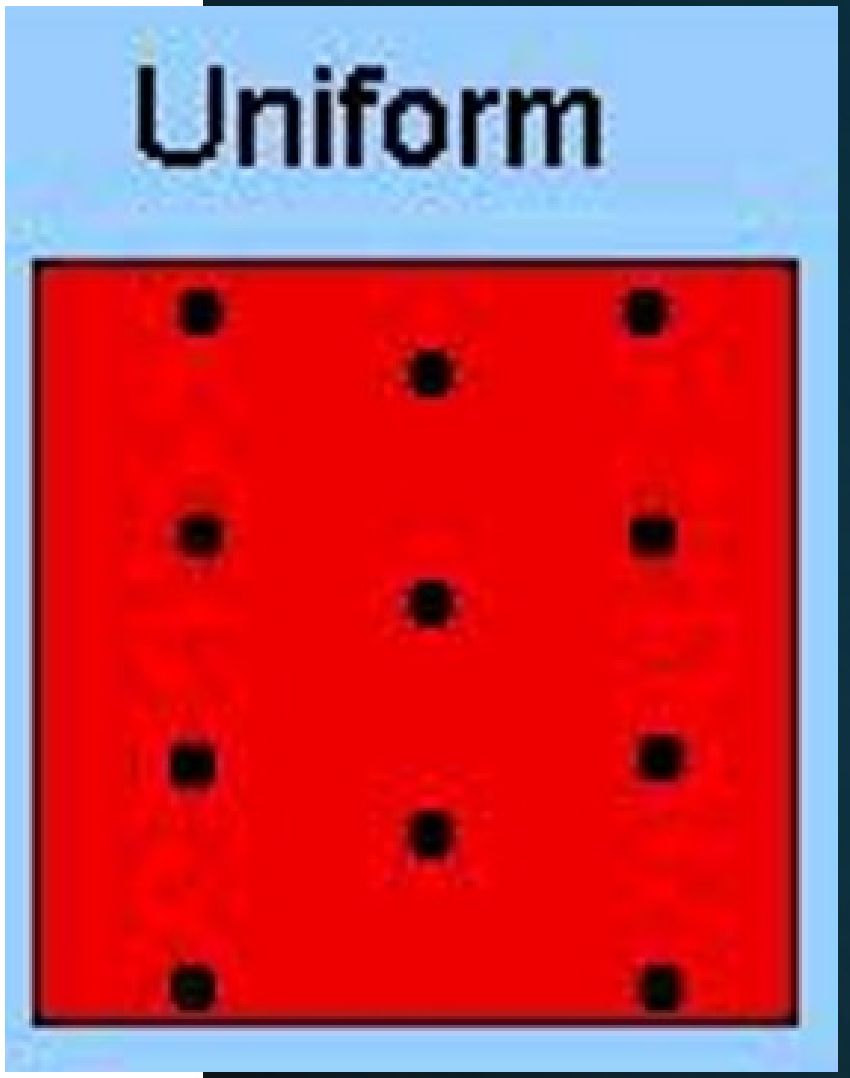


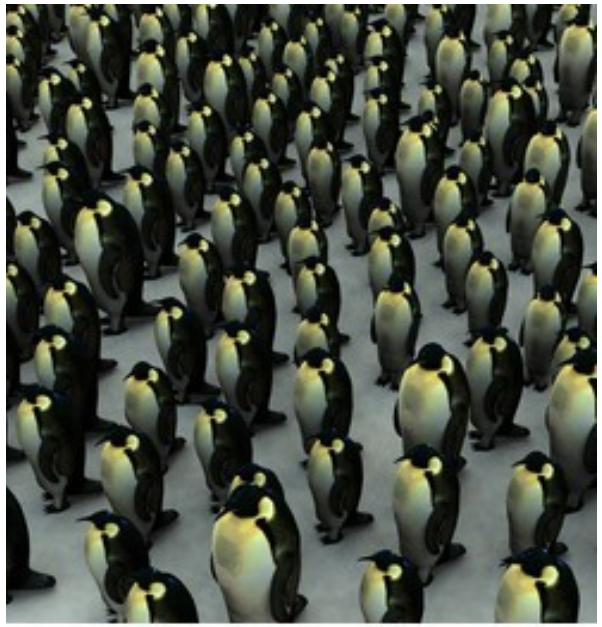
2. Uniform Distribution

- Individuals are **evenly spaced** from each other.

- **Why did it happen?**
  - Competition for resources.
  - Territorial behavior.

Spacing is regular because individuals repel each other.











Examples:

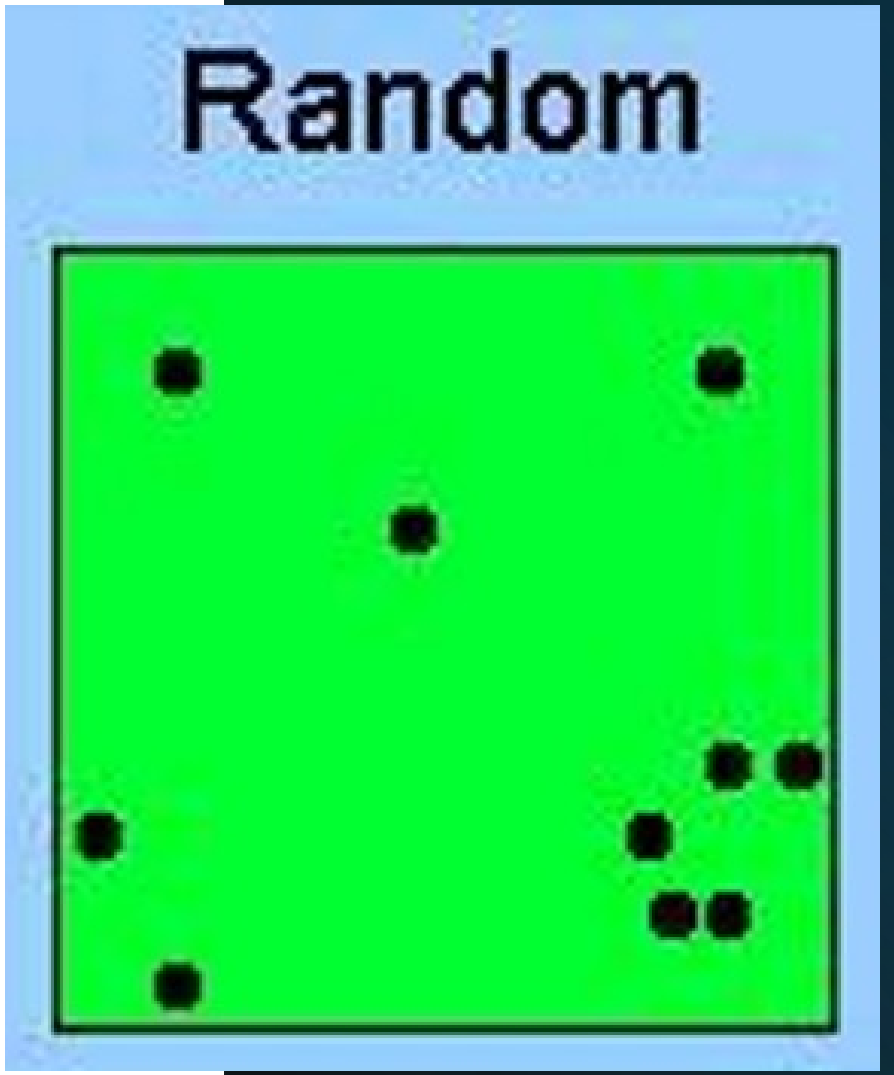
- **Penguins** nesting at equal distances.
- **Birds** defending territories.
- Trees in an **orchard plantation** (planted by humans in rows).
- **Desert plants** (like cacti) are spaced apart to avoid competition for water.

3. Random Distribution

- Individuals are spread **randomly** with no predictable pattern.

- **Why did it happen?**
  - Environment is uniform.
  - Individuals do not strongly attract or repel each other.

The **least** common type in nature.









Examples

- **Dandelions** scattered by the wind.
- Some insects are in a habitat with evenly distributed resources.
- **Wildflowers** growing randomly in a field.

Population Dynamics

- Population dynamics refer to **how and why populations change over time.**

Factors Affecting Population Change

1. **Birth Rate (Natality)** – number of individuals **born.**
2. **Death Rate (Mortality)** – number of individuals that **die.**
3. **Immigration** – individuals **entering** a population.
4. **Emigration** – individuals **leaving** a population

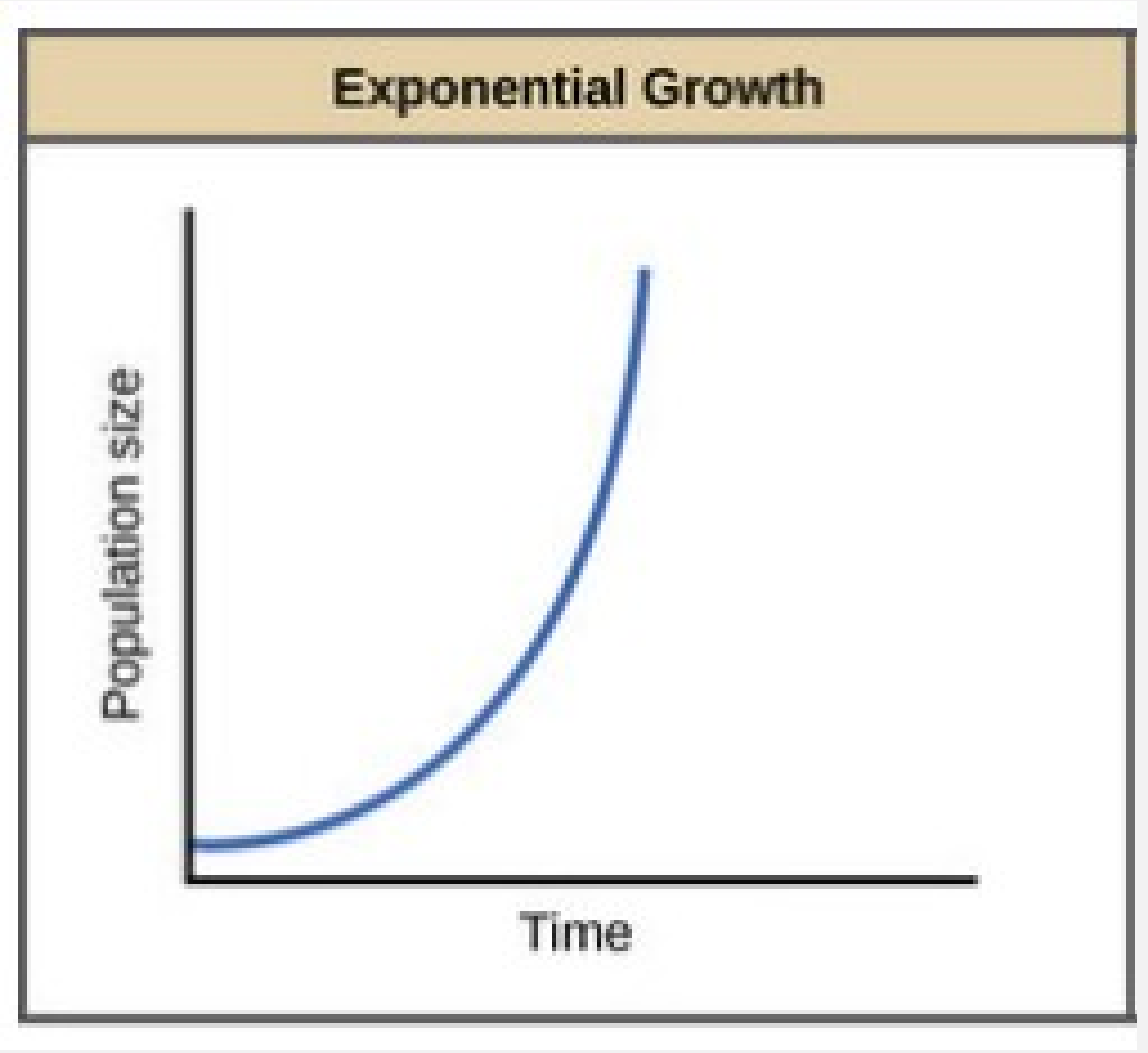
Formula:

- $Population\ Change = (Births + Immigration) - (Deaths + Emigration)$

Patterns of Population Growth

1. Exponential Growth

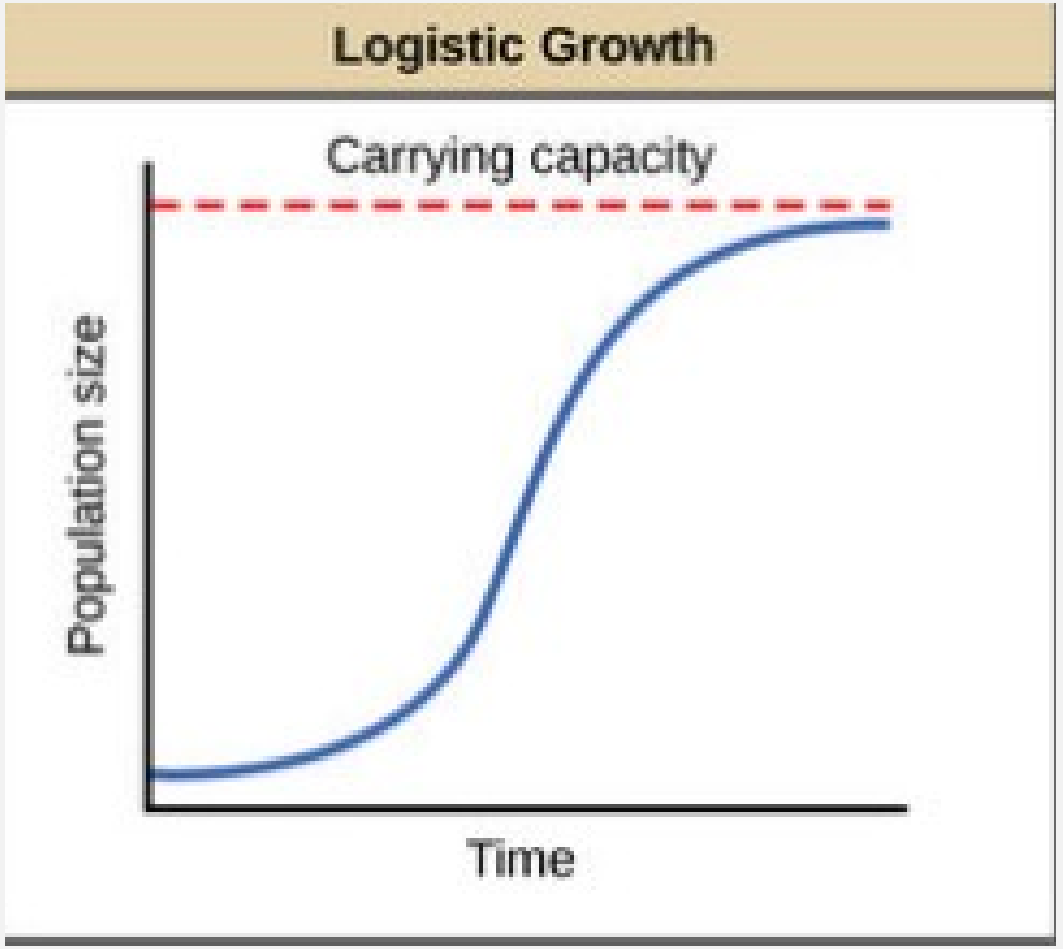
- Occurs when **resources are unlimited.**
- Population grows rapidly.
- Graph: **J-shaped curve.**
- **Example: Bacterial reproduction**
  - A bacterial population doubles every hour under ideal conditions.



Patterns of Population Growth

2. Logistic Growth

- Growth slows as **resources become limited.**
  - When a population is **small**, resources are abundant → rapid growth.
  - As population **increases**, resources become scarce → growth slows.
  - Eventually, the population **stabilizes** at **carrying capacity (K).**
- Graph: **S-shaped curve.**
- **Example**: Deer population in a forest
  - Growth slows when food, water, and space become scarce.



Carrying Capacity (K)

Carrying capacity refers to the **maximum number of individuals** of a species that **an environment can sustainably suppor**t over time.

It depends on the availability of:

- Food
- Water
- Shelter
- Space



stable level = **Carrying Capacity (K)**

Examples

1. Elephants in a Safari Park

- A park can only support **200 elephants** due to food and water.
- If elephants increase to 300 → food shortage occurs → some die or migrate.

Carrying Capacity = 200

2. Fish in a Pond

- A pond can support **500 fish**.
- If fish exceed this number → oxygen decreases → fish die.

Carrying Capacity = 500

What Happens When K is Exceeded?

If a population goes **beyond its carrying capacity**:

1. **Resource depletion occurs**
   - Food and water become insufficient.
2. **Increased competition**
   - Organisms fight for limited resources.
3. **Population crash**
   - Sudden decrease due to death, disease, or migration.

Limiting factors

- **Limiting factors** are **conditions or resources** that **restrict the growth, size, or distribution** of a population.

They **prevent populations from growing indefinitely** and help **maintain balance in ecosystems.**

No population can grow forever because:

- Resources are **limited**
- The environment can only support a certain number of individuals.

Limiting factors are what **control population size** and lead to the **carrying capacity (K)**.

Types of Limiting Factors

1. Density-Dependent Factors

These factors **depend on population density.**

- The **more crowded** the population → the **stronger the effect**.

As the population increases:

- **Disease** spreads faster.
  - Flu spreading in crowded classrooms.
- **Competition** increases.
  - Plants competing for sunlight in a forest.
- **Predators** find prey more easily.
  - Wolves hunting large herds of deer.

Types of Limiting Factors

2. Density-Independent Factors

- Affect populations **regardless of density.**
- They can impact **small or large populations equally.**

Examples:

a. Natural Disasters

- Typhoons, earthquakes, floods, and volcanic eruptions

b. Climate / Weather

- Extreme heat, drought, cold temperatures

c. Human Activities

- Pollution, deforestation, urbanization

SUMMARY TABLE

	Meaning	**Example**
Population Size	Total number of individuals.	500 fish in a pond
Population Density	Number per unit area.	50 trees per hectare
Population Distribution	How individuals are spaced.	
Clumped	Grouped together.	Herds of animals
Uniform	Evenly spaced.	Penguins nesting
Random	No pattern.	Wildflowers

Summary

- **Exponential Growth**: Rapid growth that increases continuously because **resources are unlimited**.
- **Logistic Growth**: Growth that starts rapidly but slows and stabilizes as **resources become limited.**
- Carrying Capacity (K) : **maximum sustainable population.**
  - Determined by **limited resources.**
  - Leads to a **stable population size.**
  - Can **change over time.**
  - Exceeding K causes **population decline.**
- Limiting factors: **anything that slows or stops population growth**

  Two types:
  - **Density-dependent** (depends on crowding)
  - **Density-independent** (does not depend on crowding)
  - They help determine **carrying capacity (K)**


HUMAN POPULATIONS

Human Population Growth

The human population has grown rapidly due to:

- Advances in medicine.
- Improved sanitation.
- Increased food production.

Result: **Population explosion**

AGE STRUCTURE

- **Age structure** refers to the **distribution of individuals in a population according to different age groups**.
- It shows how many people fall into each **age category.**

Age Groups

1. Pre-reproductive (0–14 years)

- Too young to reproduce.
- Future potential parents.

Example: children and teenagers.

2. Reproductive (15–44 years)

- Capable of having offspring.
- Main contributors to population growth.

Example: young adults and middle-aged individuals.

3. Post-reproductive (45+ years)

- Older individuals, typically past reproductive age.

Example: senior citizens.

Age structure diagrams

Age structure is often shown using **population pyramids**.

1. Expanding Population **(Pyramid Shape)**
2. Stable Population **(Rectangular Shape)**
3. Declining Population **(Inverted Pyramid)**

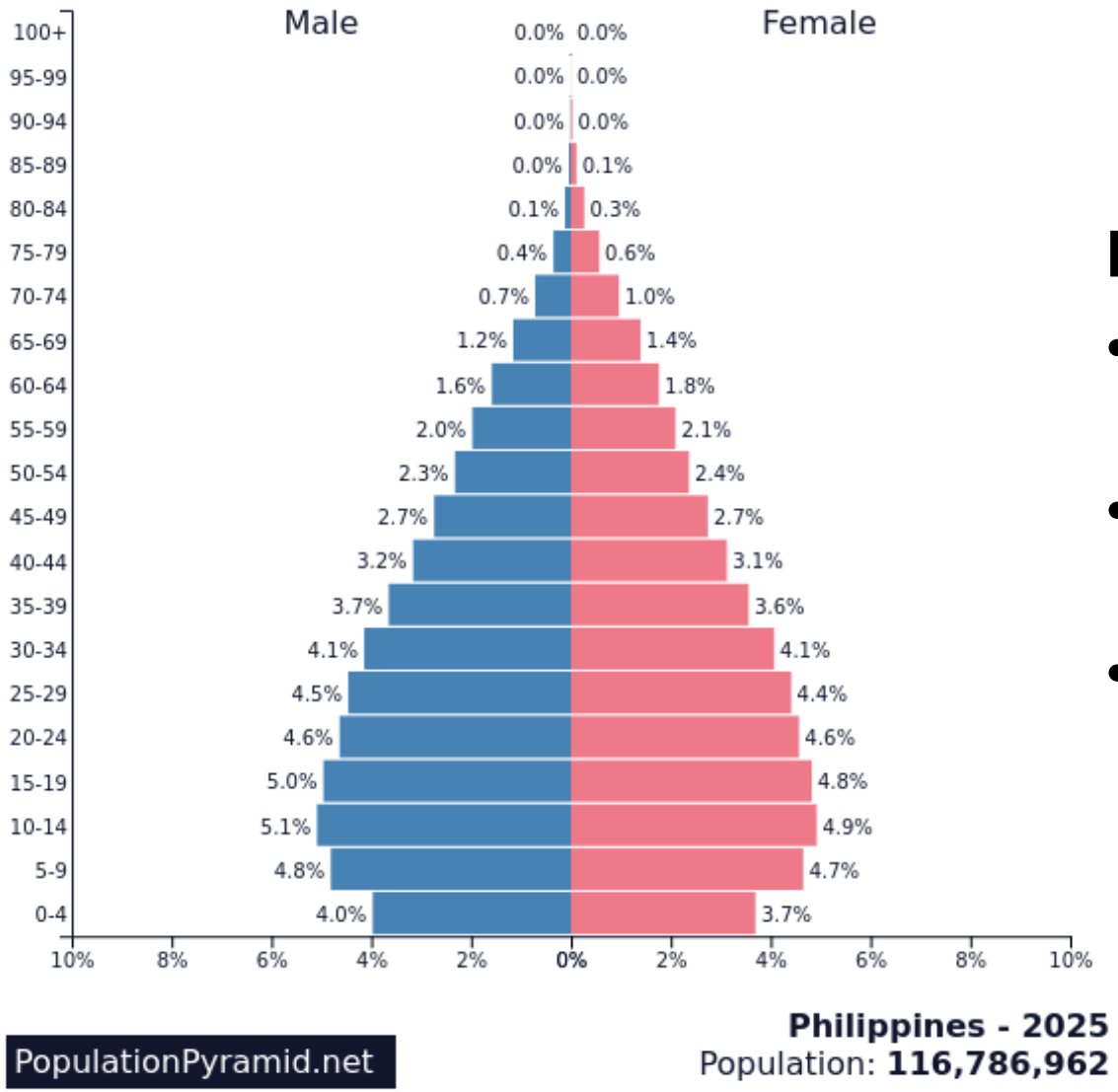
Expanding Population (Pyramid Shape)

The population **will continue to grow quickly.**

Characteristics:

- Wide base (many young people)
- High birth rate
- Rapid population growth

Examples: Philippines, Many developing countries



Philippines

- Wide base population pyramid.
- Many young people

- Result: Rapid population growth

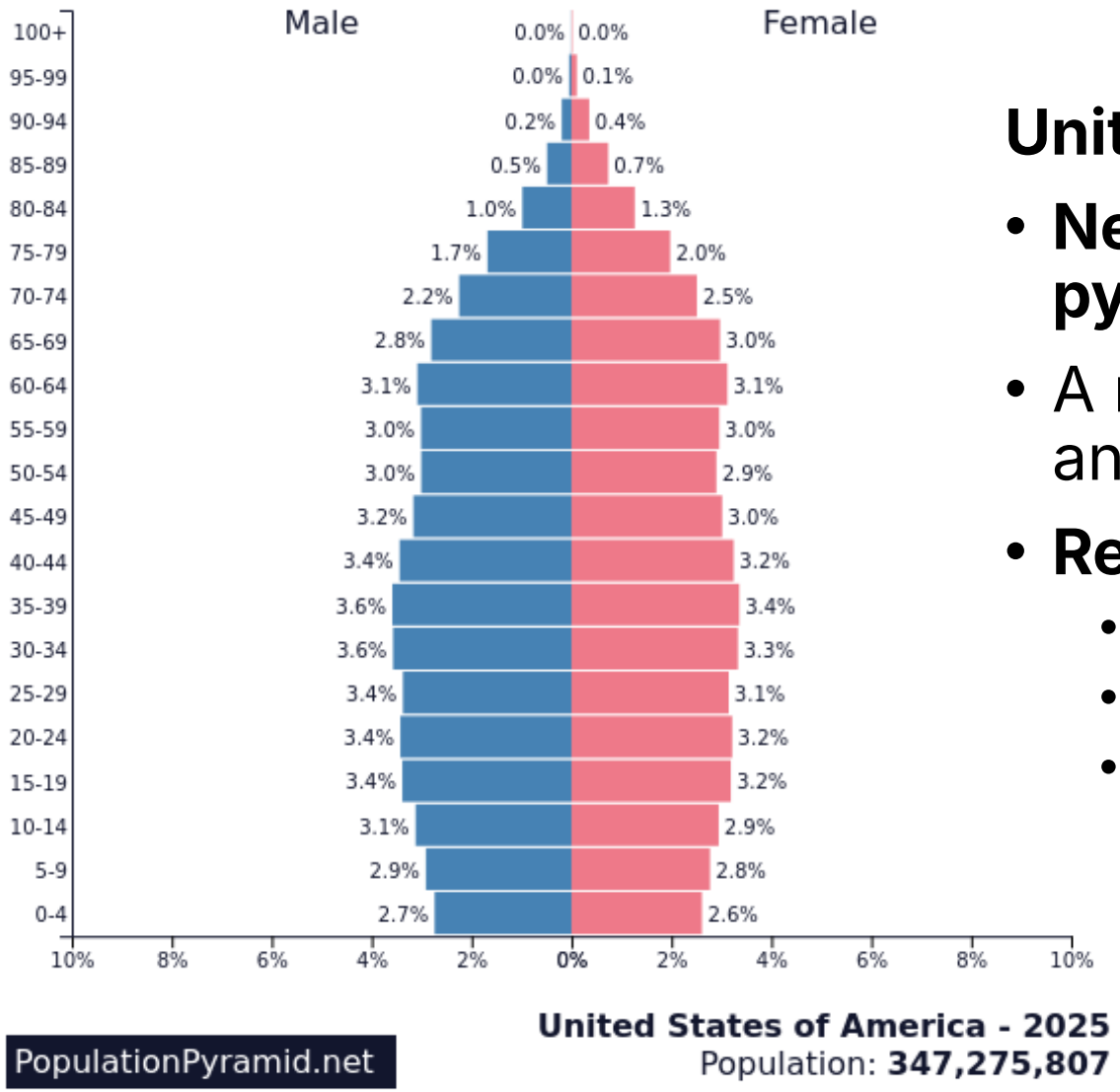
2. Stable Population (Rectangular Shape)

- Population **remains steady.**

Characteristics:

- Nearly equal numbers in all age groups
- Low birth and death rates
- Slow or no growth

Examples: USA, France



United States (USA)

- **Nearly rectangular population pyramid.**
- A moderate number of young people and adults.
- **Result:**
  - Relatively **stable population growth.**
  - Balanced working-age population.
  - Gradual aging trend (increasing elderly population).

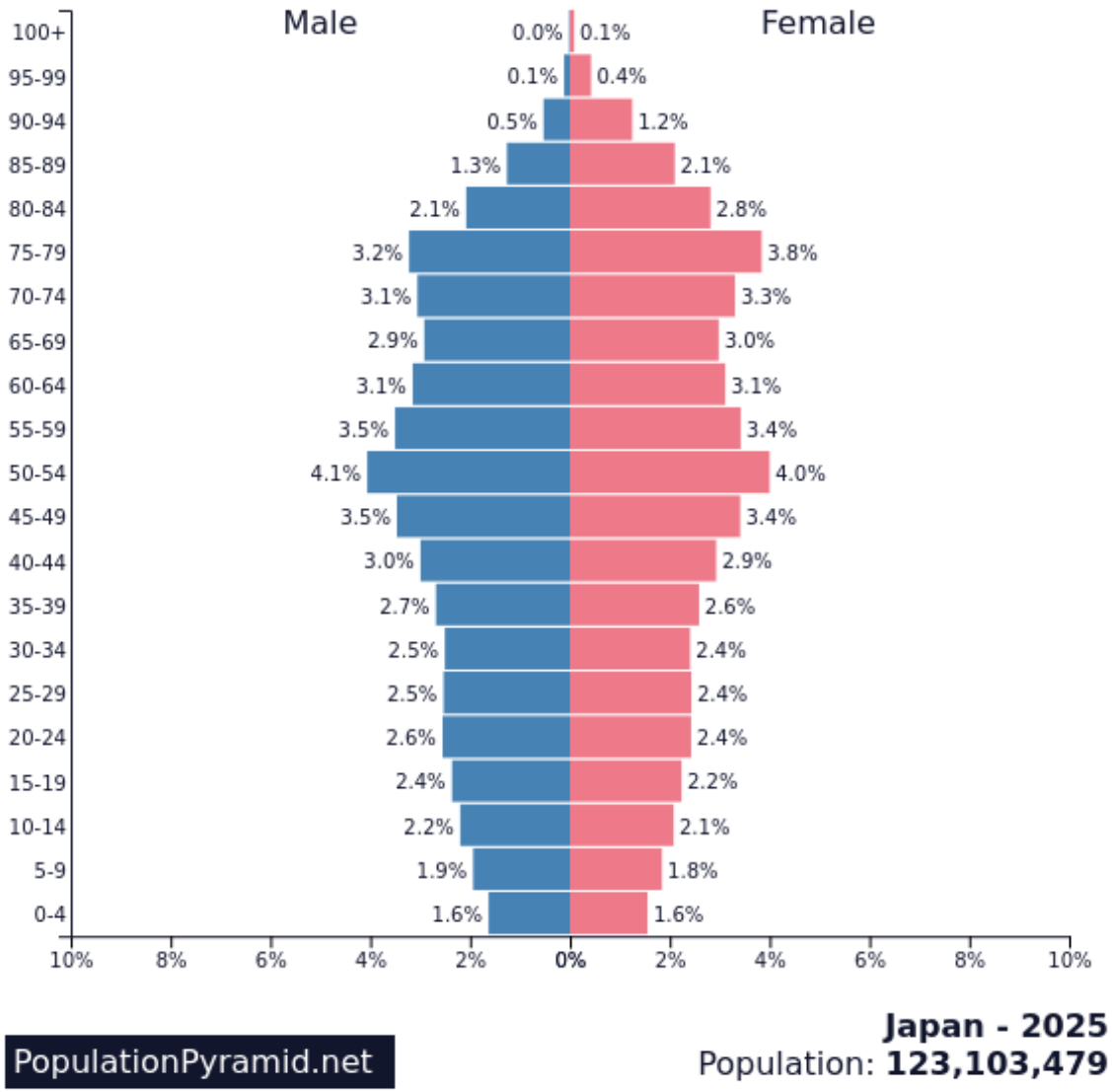
3. Declining Population (Inverted Pyramid)

- Population may **shrink** over time.

Characteristics:

- Narrow base (few young people)
- Low birth rate
- Aging population

Examples: Japan, Germany



Japan

- Narrow base, more elderly
- Result:
  - Aging society
  - Labor shortage

Why age structure is important?

1. **Predicts Future Growth**
   - Many young people → population will grow
   - Few young people → population may decline
2. **Helps Government Planning**
   - More children → need more schools
   - More elderly → need more healthcare services
3. **Shows Economic Impact**
   - Large working population → strong economy
   - Aging population → fewer workers

Factors affecting age structure

1) Birth rate
2) Death rate
3) Health care
4) Culture and traditions
5) Government policies

LIFE EXPECTANCY

- **Life expectancy** is the **average number of years a person is expected to live** in a population.
- It reflects the **overall health and living conditions** of a country.
- It is influenced by:
  - Healthcare quality
  - Nutrition
  - Sanitation
  - Lifestyle
- **High** life expectancy → better quality of life.
- **Low** life expectancy → possible health or economic problems.

Philippines

- Around **low to moderate life expectancy.**
- roughly **72.04 years**. about 66.89 years for men and 72.82 years for women.

Reason: Improving healthcare but still limited in some areas.

Japan

- Very **high life expectancy** (among the highest in the world).
- **85.0 years**

Reason:

- Healthy diet (fish, vegetables)
- Strong healthcare system

USA

- High but slightly lower than Japan
- **79.4 years**
- Reason: Advanced healthcare, but lifestyle diseases affect longevity.

SEX RATIO

- **Sex ratio** is the **number of males compared to females in a population**.
- Usually expressed as:
  - $Sex\ Ratio = \frac{Number\ of\ Males}{Number\ of\ Females}$
- **Shows gender balance** in a population. Important for predicting:
  - Marriage patterns
  - Workforce distribution
  - Social dynamics

Imbalance may lead to:

- Social issues
- Marriage imbalance
- Labor shortages/surpluses

Cont.

Global Average

- About **105 males for every 100 females at birth.**

Philippines

- Almost balanced sex ratio (99 to 102 males per 100 females).

USA

- Slightly **more females than males.**
- Reason: Females generally live longer.

India / China (some regions)

- More males than females.
- Reason: cultural preferences (historically).

FERTILITY RATE

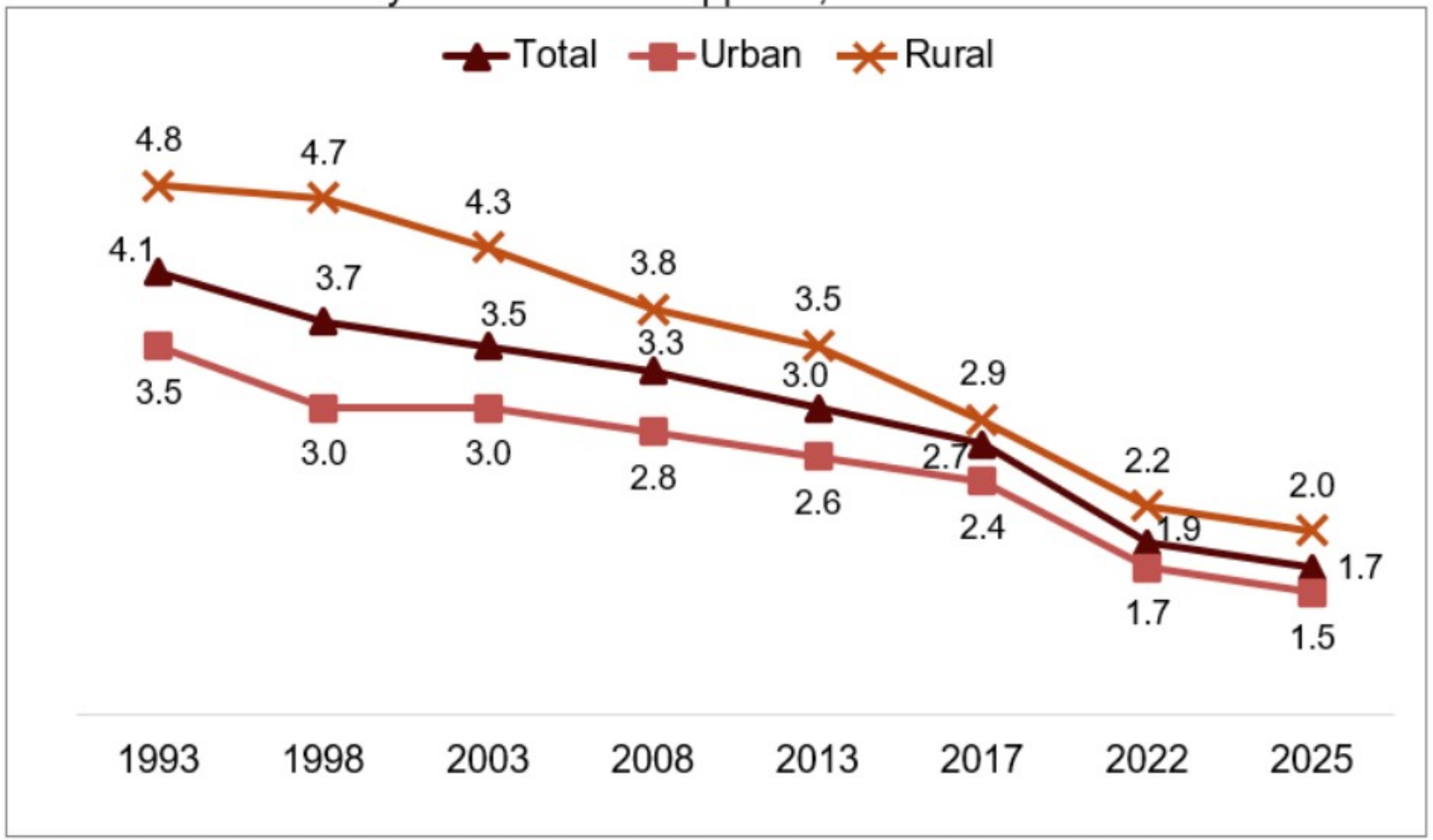
- **Total Fertility Rate (TFR)** is the average number of children a woman would have by the end of her childbearing years if she were to bear children at the current age-specific fertility rates.

- **Replacement level** = about **2.1 children per woman.**
  - meaning the population can replace itself from one generation to the next.

  - **Above 2.1** → population grows
  - **Below 2.1** → population declines

- The Philippines' **TFR** dropped to **1.7** children per woman in 2025.

Figure 1. Total Fertility Rates for the Three Years Preceding the Survey by Residence: Philippines, 1993 to 2025



Source: Philippine Statistics Authority, 1993 to 2025 National Demographic and Health Survey

Factors Affecting Fertility Rate

- Education (especially of women).
- Access to family planning.
- Economic status.
- Cultural beliefs.
- Government policies.

Impacts of Human Population Growth

Positive Effects

- Economic development.
- Technological advancement.

Negative Effects

- Environmental degradation
- Resource depletion
- Pollution
- Climate change

Population and the Environment

Population growth affects ecosystems through:

- Overuse of natural resources
- Habitat destruction
- Loss of biodiversity

Summary

Age structure = **age distribution of a population.**

- Three groups:
  - Pre-reproductive
  - Reproductive
  - Post-reproductive
- Shapes tell:
  - Growing population → pyramid
  - Stable population → rectangular
  - Declining population → inverted

Summary

	Meaning	**High Value Means**	**Low Value Means**
Life Expectancy	Average lifespan	Better health, longer life	Poor health conditions
Sex Ratio	Males vs females	Possible imbalance	Balanced population
Fertility Rate	Average number of children per woman.	Population growth	Population decline

Summary

- **Age structure**
- **Life expectancy**
- **Sex ratio**
- **Fertility rate**

These help us understand:

- Population growth
- Aging society
- Future population trends